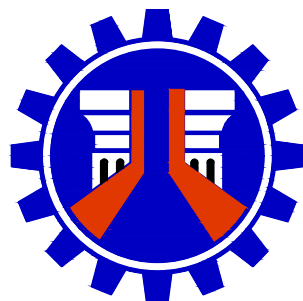




REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGIONAL OFFICE I
SAN FERNANDO CITY, LA UNION

CY 2024 PROJECT
DETAILED ENGINEERING DESIGN PLAN FOR
**CONSTRUCTION/REHABILITATION OF FLOOD MITIGATION FACILITIES WITHIN MAJOR RIVER BASINS AND
PRINCIPAL RIVERS, CONSTRUCTION OF FLOOD CONTROL STRUCTURE ALONG ARINGAY RIVER,
BARANGAY HALOG EAST, TUBAO, LA UNION**
LOCATION: TUBAO, LA UNION
NET LENGTH: 675.00 L.M.

SUBMITTED:

ALBERTO B. ESQUEJO, JR.
CHIEF, PLANNING AND DESIGN DIVISION

DATE:

RECOMMENDED:

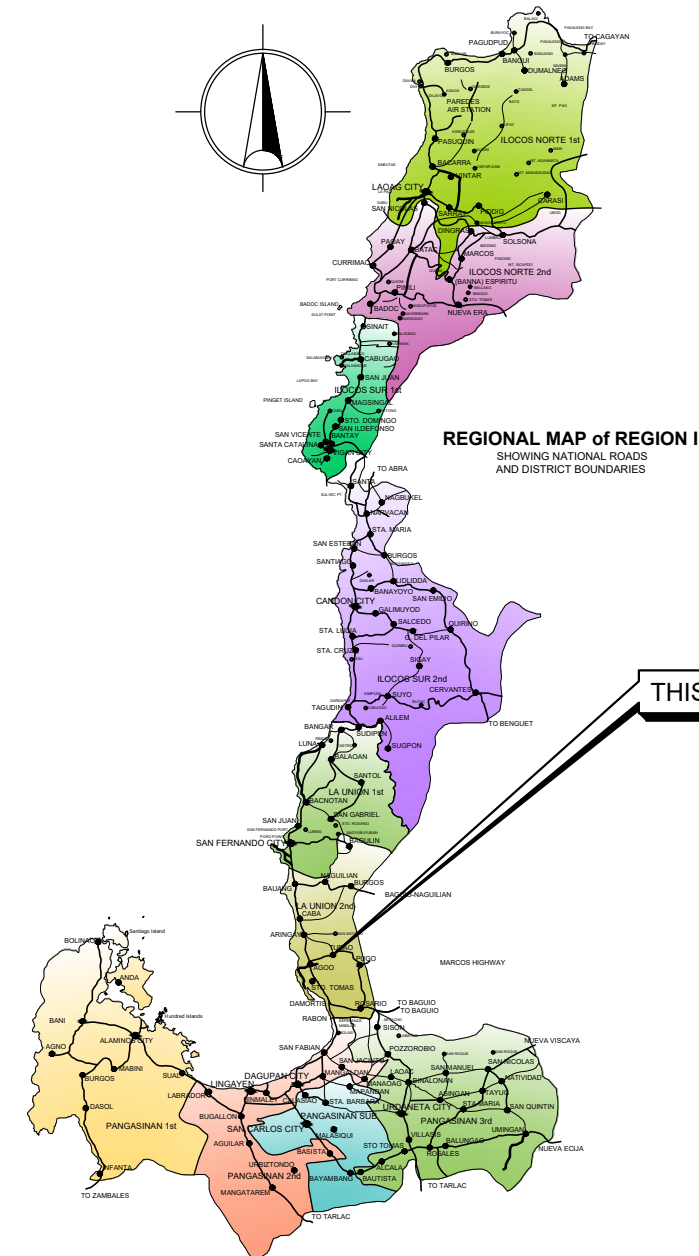
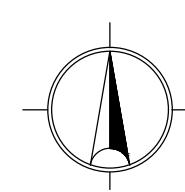
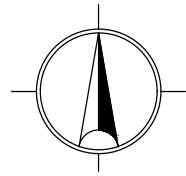
MATHIAS L. MALENAB
ASSISTANT REGIONAL DIRECTOR

DATE:

APPROVED:

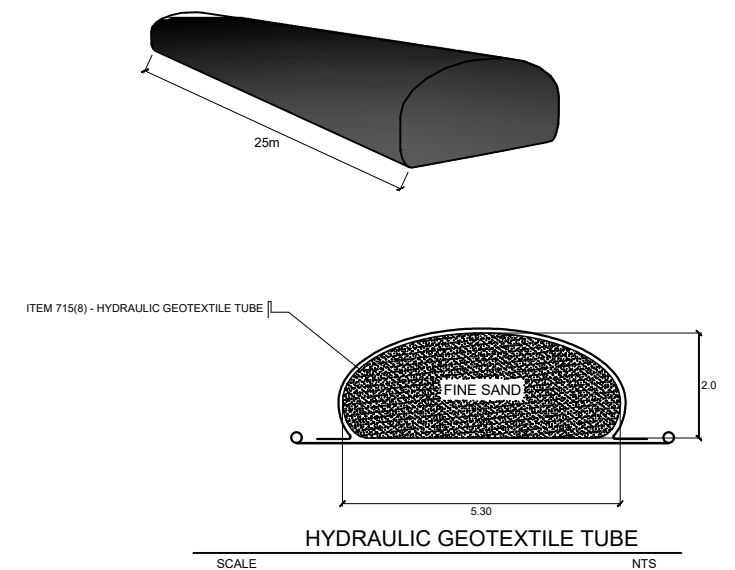
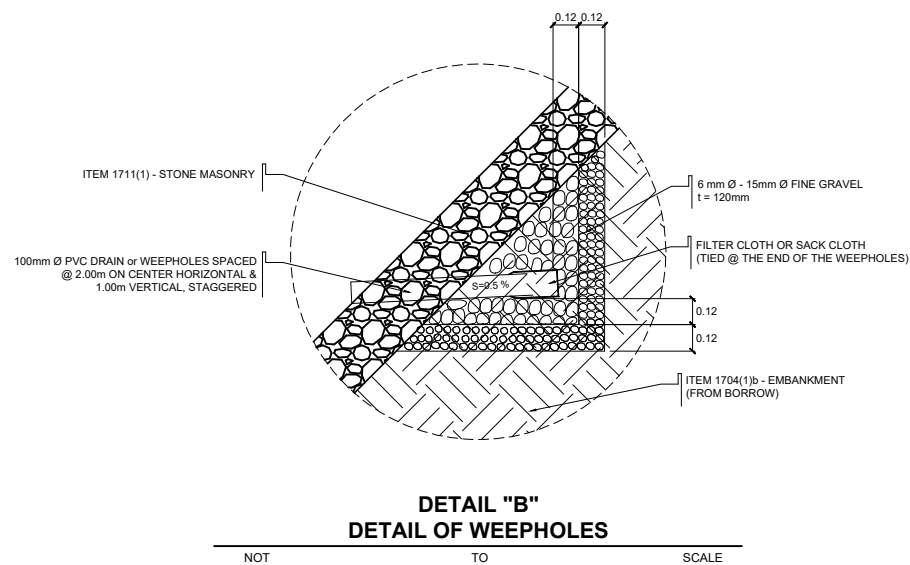
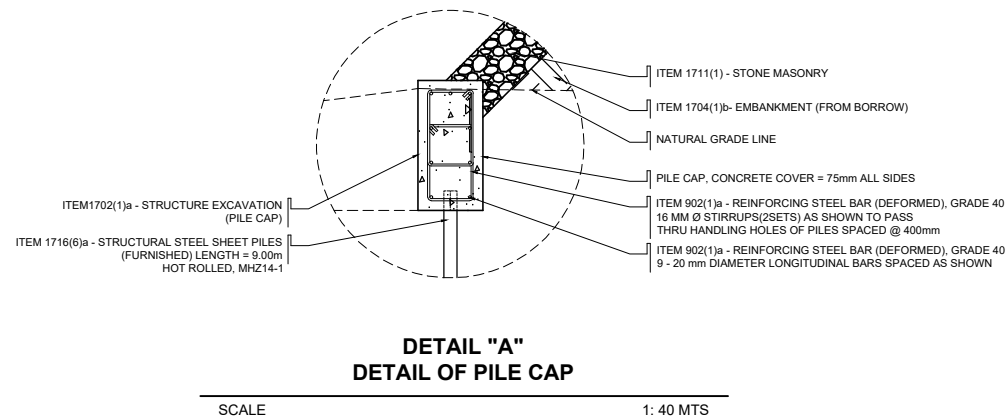
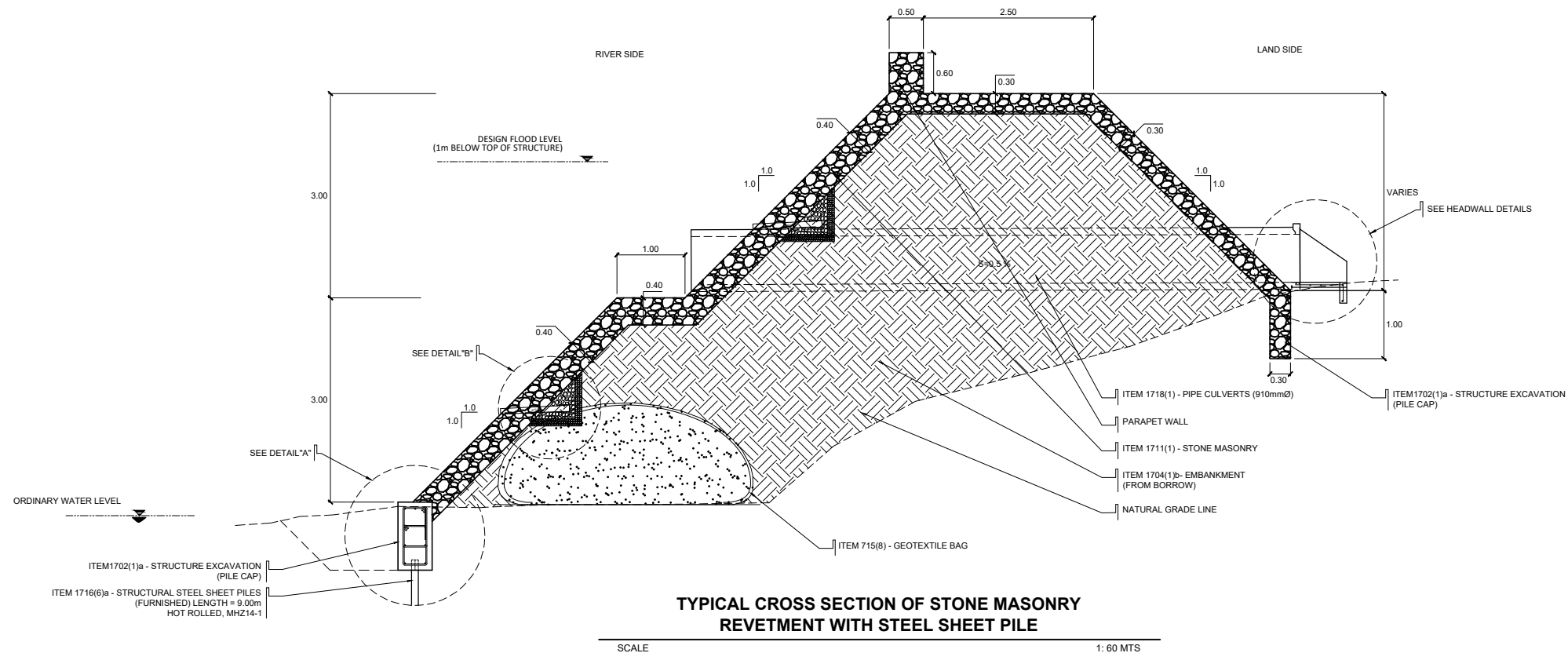
RONNEL M. TAN, CESO III
REGIONAL DIRECTOR

DATE:



 REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGIONAL OFFICE I SAN FERNANDO CITY, LA UNION	PROJECT NAME AND LOCATION: CONSTRUCTION/REHABILITATION OF FLOOD MITIGATION FACILITIES WITHIN MAJOR RIVER BASINS AND PRINCIPAL RIVERS. CONSTRUCTION OF FLOOD CONTROL STRUCTURE ALONG ARINGAY RIVER, BARANGAY HALOG EAST, TUBAO, LA UNION TUBAO, LA UNION	SHEET CONTENTS: VICINITY MAP LOCATION MAP	DRAFTED: AMIEL MAUREEN V. RAMOS ENGINEER II	REVIEWED: MARIA MONINA P. DELIZO OIC, FLOOD CONTROL, SOCIAL AND ENVIRONMENTAL SECTION DATE:	SUBMITTED: ALBERTO B. ESQUEJO, JR. CHIEF, PLANNING AND DESIGN DIVISION DATE:	RECOMMENDED: MATHIAS L. MALENAB ASST. REGIONAL DIRECTOR DATE:	APPROVED: RONNEL M. TAN, CESO III REGIONAL DIRECTOR DATE:	SET NO.: 	SHEET NO.: 
			PREPARED: JHAMALI C. PANGCATAN ENGINEER II						

 REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGIONAL OFFICE I SAN FERNANDO CITY, LA UNION	PROJECT NAME AND LOCATION:	SHEET CONTENTS:	DRAFTED:	REVIEWED:	SUBMITTED:	RECOMMENDED:	APPROVED:	SET NO.:	SHEET NO.:
	CONSTRUCTION/REHABILITATION OF FLOOD MITIGATION FACILITIES WITHIN MAJOR RIVER BASINS AND PRINCIPAL RIVERS, CONSTRUCTION OF FLOOD CONTROL STRUCTURE ALONG ARINGAY RIVER, BARANGAY HALOG EAST, TUBAO, LA UNION TUBAO, LA UNION	GENERAL NOTES SUMMARY OF QUANTITIES	AMIEL MAUREEN V. RAMOS ENGINEER II PREPARED: JHAMALI C. PANGCATAN ENGINEER II	MARIA MONINA P. DELIZO OIC, FLOOD CONTROL, SOCIAL AND ENVIRONMENTAL SECTION DATE:	ALBERTO B. ESQUEJO, JR. CHIEF, PLANNING AND DESIGN DIVISION DATE:	MATHIAS L. MALENAB ASST. REGIONAL DIRECTOR DATE:	RONNEL M. TAN, CESO III REGIONAL DIRECTOR DATE:		



REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGIONAL OFFICE I
SAN FERNANDO CITY, LA UNION

PROJECT NAME AND LOCATION:
CONSTRUCTION/REHABILITATION OF FLOOD MITIGATION FACILITIES
WITHIN MAJOR RIVER BASINS AND PRINCIPAL RIVERS,
CONSTRUCTION OF FLOOD CONTROL STRUCTURE ALONG ARINGAY
RIVER, BARANGAY HALOG EAST, TUBAO, LA UNION
TUBAO, LA UNION

SHEET CONTENTS:
TYPICAL CROSS SECTION OF STONE MASONRY
REVETMENT
PILE CAP DETAILS
WEEPHOLE DETAILS

DRAFTED:
AMIEL MAUREEN V. RAMOS
ENGINEER II
PREPARED:
JHAMALI C. PANGCATAN
ENGINEER II

REVIEWED:
MARIA MONINA P. DELIZO
OIC, FLOOD CONTROL, SOCIAL AND
ENVIRONMENTAL SECTION
DATE:

SUBMITTED:
ALBERTO B. ESQUEJO, JR.
CHIEF, PLANNING AND DESIGN DIVISION
DATE:

RECOMMENDED:
MATHIAS L. MALENAB
ASST. REGIONAL DIRECTOR
DATE:

APPROVED:
RONNEL M. TAN, CESO III
REGIONAL DIRECTOR
DATE:

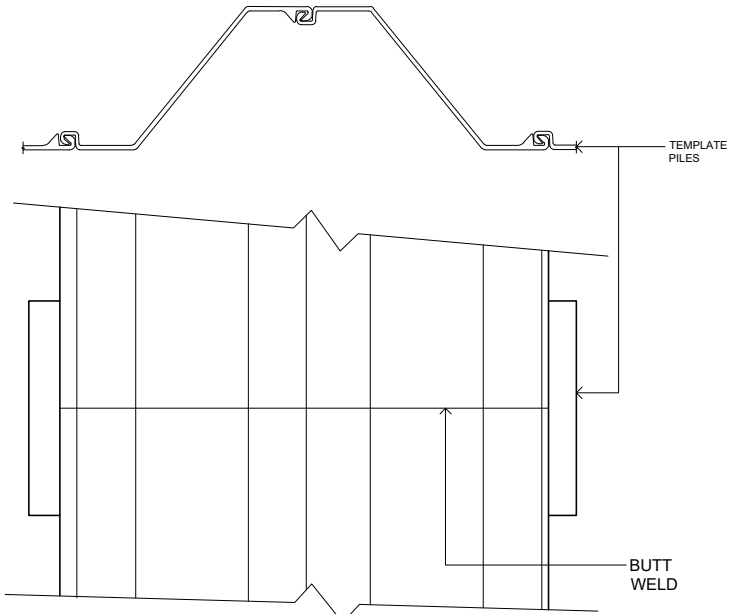
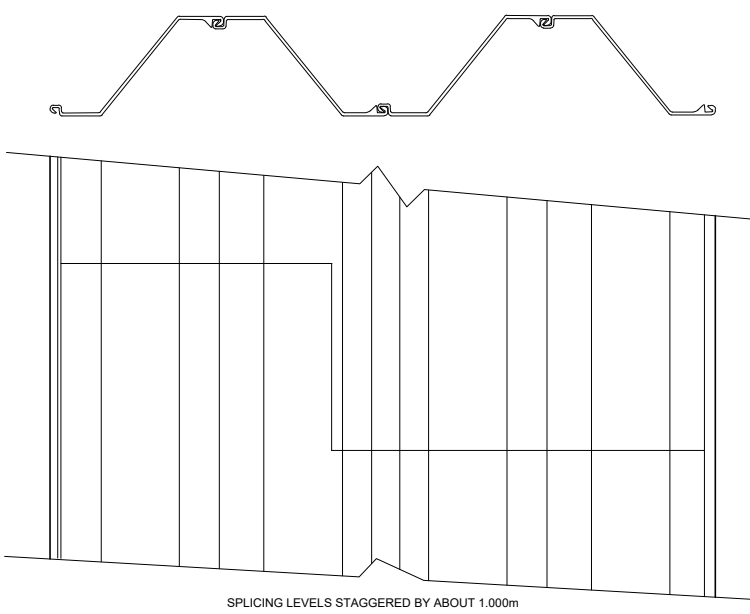
SET NO.:
SHEET NO.:
05
08

PROPERTIES OF STEEL					SHEET PILE			
TYPE	W mm (NOMINAL)	h mm (NOMINAL)	t1 mm (NOMINAL)	t2 mm (NOMINAL)	WEIGHT, Wt. kg/m	SECTIONAL AREA OF WALL cm ² /m	SECTION MODULUS cm ³ /m	MOMENT OF INERTIA cm ⁴ /cm
Z-TYPE	770	344.50	9.60	9.50	79.60	132.20	1,357	23,370
Z-SHAPE SSP HOT-ROLLED					MECHANICAL PROPERTIES: EN 10248 S355GP Fy = 355 Mpa, Fu = 480 Mpa			

SCHEDULE OF STEEL SHEET PILE JOINT		
TYPE	PL1 (STEEL PLATE)	
	B1 mm	mm
Z-SHAPE	450	14.00

STRUCTURAL PLATE: ASTM A50, Fy = 345 Mpa, Fu = 400 Mpa MILD STEEL COATED ELECTRODES: E7018-X, Fy = 400 Mpa, Fu = 490 Mpa	
NOTES: 1. STEEL BASE METAL TO BE WELDED SHALL BE OPEN-HEARTH OR ELECTRIC FURNACE STEEL CONFORMING TO AASHTO M183. 2. t1 = t or THICKER	
 BASE METAL THK LESS THAN 6mm BASE METAL 6mm OR MORE IN THICKNESS	

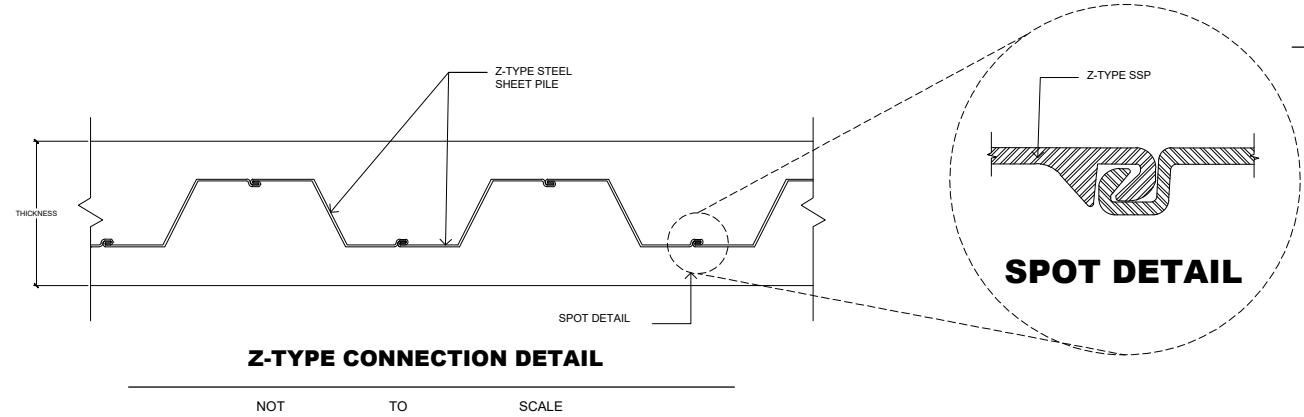
- NOTE:
- ALL WORK SHALL BE DONE IN ACCORDANCE WITH ALL PROVISIONS IN ITEM 509 OF DPWH STANDARD SPECIFICATION.
 - BEFORE ANY WORK BEGINS IN ENVIRONMENTALLY SENSITIVE AREAS, TEMPORARY EROSION CONTROL MEASURES AND PROPER MANAGEMENT PRACTICES SHALL BE IN PLACE.
 - ALL TEMPORARY FENCING FOR PROTECTED AREAS SHALL BE INSTALLED PRIOR TO COMMENCEMENT OF WORK.
 - LOW GROUND DISTURBING EQUIPMENT SHALL BE UTILIZED IN ENVIRONMENTALLY SENSITIVE AREAS AS APPROVED BY THE CIVIL ENGINEER-IN-CHARGE.
 - STEEL SHEET PILES SHALL MEET THE REQUIREMENTS OF AASHTO M 2020 (ASTM A328), OR AASHTO M 223. THE JOINTS SHALL BE PRACTICALLY WATER-TIGHT WHEN THE PILES ARE IN PLACE.
 - SHEET PILES SHALL BE DRIVEN TO ELEVATION SHOWN ON THE PLANS OR AS DIRECTED BY ENGINEER-IN-CHARGE.
 - THE REQUIREMENTS GOVERNING THE INSTALLATION OF SHEET PILING SHALL CONFORM IN GENERAL TO THOSE GOVERNING BEARING PILES AS SET FORTH UNDER ITEM 400 OF DPWH STANDARD SPECIFICATION, PILING.
 - CORROSION PROTECTION MATERIAL SHALL BE APPLIED TO STEEL SHEET PILE PRIOR TO DRIVING.



BUTT WELDING / SOUDAGE BOUT

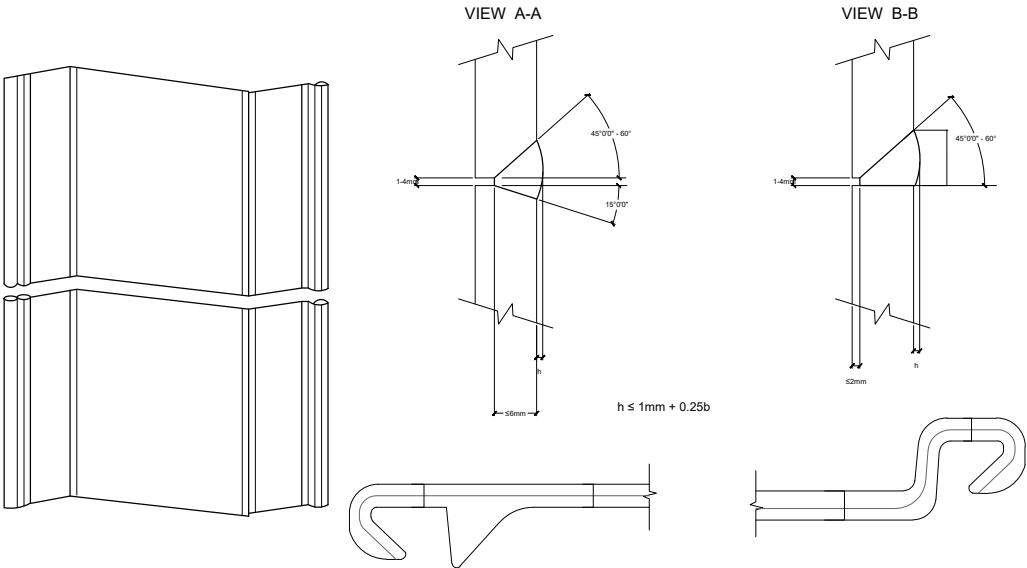
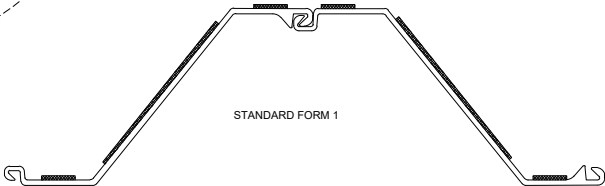
NOT TO SCALE

NOT TO SCALE



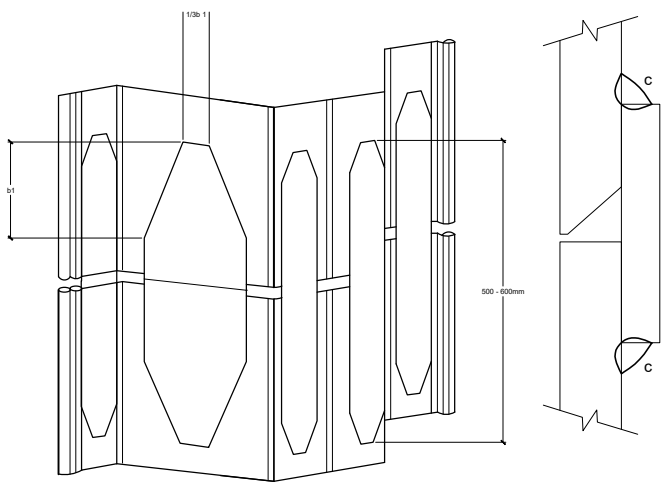
Z-TYPE CONNECTION DETAIL

NOT TO SCALE



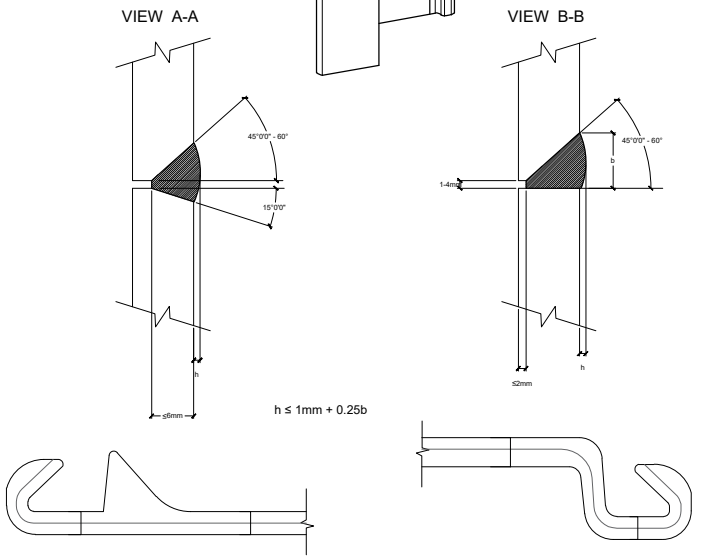
BUTT WELDING SHEET PILES

NOT TO SCALE



SPLICING OF SHEET PILES USING PLATES

NOT TO SCALE



BUTT WELDING SHEET PILES



REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGIONAL OFFICE I
SAN FERNANDO CITY, LA UNION

PROJECT NAME AND LOCATION:
CONSTRUCTION/REHABILITATION OF FLOOD MITIGATION FACILITIES
WITHIN MAJOR RIVER BASINS AND PRINCIPAL RIVERS,
CONSTRUCTION OF FLOOD CONTROL STRUCTURE ALONG ARINGAY
RIVER, BARANGAY HALOG EAST, TUBAO, LA UNION
TUBAO, LA UNION

SHEET CONTENTS:
DETAILS OF STEEL SHEET PILE
(Z-TYPE, HOT ROLLED)

DRAFTED:
AMIEL MAUREEN V. RAMOS
ENGINEER II
PREPARED:
JHAMALI C. PANGCATAN
ENGINEER II

REVIEWED:
MARIA MONINA P. DELIZO
OIC, FLOOD CONTROL, SOCIAL AND
ENVIRONMENTAL SECTION
DATE:

SUBMITTED:
ALBERTO B. ESQUEJO, JR.
CHIEF, PLANNING AND DESIGN DIVISION
DATE:

RECOMMENDED:
MATHIAS L. MALENAB
ASST. REGIONAL DIRECTOR
DATE:

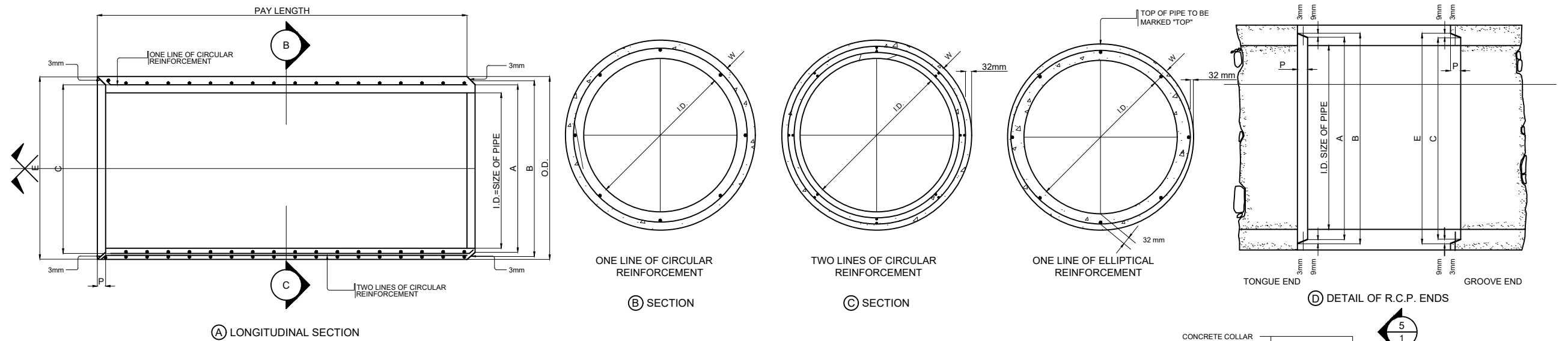
APPROVED:
RONNEL M. TAN, CESO III
REGIONAL DIRECTOR
DATE:

SET NO.:



SHEET NO.:



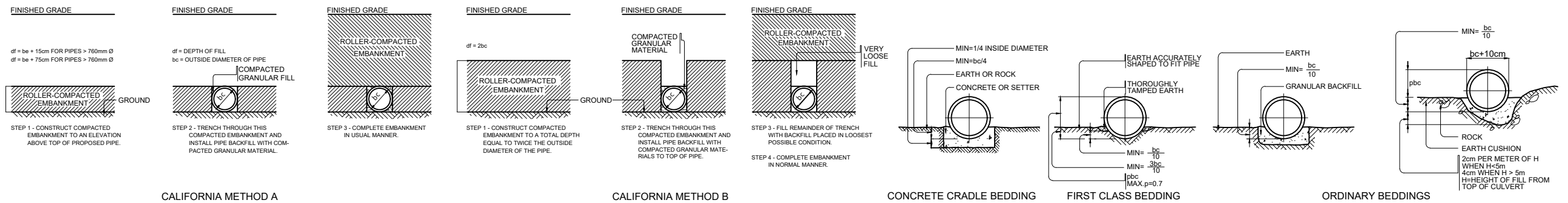
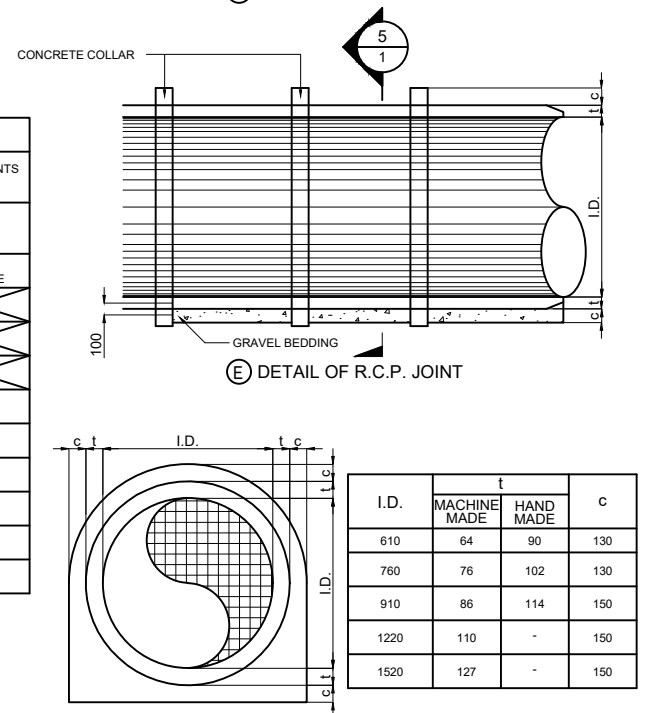


		STANDARD STRENGTH REINFORCED CONCRETE PIPE CULVERTS														EXTRA STRENGTH REINFORCED CONCRETE PIPE CULVERTS													
		CONCRETE 247 kg/cr ³ (3,500 lb/in) ²								CONCRETE 317 kg/cr ³ (4,500 lb/in) ²								STRENGTH TEST REQUIREMENTS kg/m OF PIPE		CONCRETE 317 kg/cr ³ (4,500 lb/in) ²								STRENGTH TEST REQUIREMENTS kg/m OF PIPE	
SIZE OF PIPE		WALL THICK-NESS (mm)	TONGUE (mm)		GROOVE (mm)		DEPTH (mm)	MINIMUM REINFORCEMENT cfr /m OF PIPE		WALL THICK-NESS (mm)	TONGUE (mm)		GROOVE (mm)		DEPTH (mm)	MINIMUM REINFORCEMENT cfr /m OF PIPE		THREE-EDGE-BEARING METHOD ✱		WALL THICK-NESS (mm)	TONGUE (mm)		GROOVE (mm)		DEPTH (mm)	MINIMUM REINFORCEMENT cfr /m OF PIPE		THREE-EDGE-BEARING METHOD	
(in)	(mm)		CIRCULAR REINFORCEMENT	ELLIPTICAL REINFORCEMENT	CIRCULAR REINFORCEMENT	ELLIPTICAL REINFORCEMENT		0.00025mCRACK LOAD	ULTIMATE LOAD		CIRCULAR REINFORCEMENT	ELLIPTICAL REINFORCEMENT	0.00025mCRACK LOAD	LOAD ULTIMATE															
I.D.	I.D.	W	A	B	C	E	P	CIRCULAR REINFORCEMENT	ELLIPTICAL REINFORCEMENT	W	A	B	C	E	P	CIRCULAR REINFORCEMENT	ELLIPTICAL REINFORCEMENT	0.00025mCRACK LOAD	ULTIMATE LOAD	W	A	B	C	E	P	CIRCULAR REINFORCEMENT	ELLIPTICAL REINFORCEMENT	0.00025mCRACK LOAD	LOAD ULTIMATE
12	300	57	344	363	351	370	44	1 LINE 1.48		51	495	514	502	521	44	1 LINE 1.69		3.355	5.218	-	-	-	-	-					
15	380	57	344	363	351	370	44	1 LINE 1.90		51	495	514	502	521	44	1 LINE 2.33		3.914	6.060	-	-	-	-	-					
18	460	64	508	527	514	534	44	1 LINE 2.54	1 LINE 2.12	51	495	514	502	521	44	1 LINE 2.96		4.473	6.709	-	-	-	-	-					
24	610	76	673	692	680	699	44	1 LINE 3.60	1 LINE 2.75	64	660	680	667	686	44	1 LINE 4.23	1 LINE 3.60	4.473	7.454	76	673	692	680	699	44	1 LINE 5.50	1 LINE 4.23	5.964	8.945
30	760	89	858	857	845	864	51	1 LINE 4.66	1 LINE 3.60	76	825	845	832	851	51	1 LINE 5.92	1 LINE 4.44	5.032	8.573	89	838	857	845	864	51	1 LINE 6.56	1 LINE 5.08	7.454	11.182
36	910	102	1003	1022	1010	1029	64	2 LINES EACH 3.81	1 LINE 3.81	86	988	1007	994	1013	64	2 LINES EACH 4.66	1 LINE 4.66	6.038	9.840	102	1003	1022	1010	1029	64	2 LINES EACH 5.92	1 LINE 5.92	8.945	13.418
42	1070	114	1168	1187	1175	1194	64	2 LINES EACH 4.44	1 LINE 4.44	95	1150	1165	1156	1175	64	2 LINES EACH 5.29	1 LINE 5.29	7.045	10.958	114	1168	1187	1175	1194	64	2 LINES EACH 6.98	1 LINE 6.98	10.436	15.655
48	1220	127	1334	1353	1340	1359	64	2 LINES EACH 5.29	1 LINE 5.29	108	1315	1334	1321	1340	64	2 LINES EACH 6.56	1 LINE 6.56	8.051	11.927	127	1334	1353	1340	1359	64	2 LINES EACH 8.04	1 LINE 8.04	11.927	17.891
60	1520	152	1664	1683	1670	1690	64	2 LINES EACH 6.98	1 LINE 6.98	127	1639	1658	1645	1664	64	2 LINES EACH 8.68	1 LINE 8.68	8.945	14.909	152	1664	1683	1670	1690	64	2 LINES EACH 10.58	1 LINE 10.58	13.418	22.364

⊗ THE DISTANCE FROM CENTERLINE OF THE REINFORCEMENT TO THE NEAREST SURFACE OF THE CONCRETE HAS BEEN ASSUMED AS 32mm FOR PIPES WITH A SHELL THICKNESS OF 64mm OR MORE.

* TEST LOADS FOR SAND-BEARING TEST SHALL BE ONE AND ONE - HALF TIMES THOSE SPECIFIED IN THIS TABLE FOR THE THREE - EDGE BEARING TEST.

NOTE : THIS DRAWING WAS COPIED FROM THE PREVIOUSLY APPROVED CARMONA - TERNATE - NASUGBU - LOOC PROJECT.



REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGIONAL OFFICE I
SAN FERNANDO CITY, LA UNION

PROJECT NAME AND LOCATION:
CONSTRUCTION/REHABILITATION OF FLOOD MITIGATION FACILITIES
WITHIN MAJOR RIVER BASINS AND PRINCIPAL RIVERS,
CONSTRUCTION OF FLOOD CONTROL STRUCTURE ALONG ARINGAY
RIVER, BARANGAY HALOG EAST, TUBAO, LA UNION
TUBAO, LA UNION

SHEET CONTENTS:
STANDARD REINFORCED CONCRETE PIPE
CULVERT (RCP) DETAILS

DRAFTED:
AMIEL MAUREEN V. RAMOS
ENGINEER II
PREPARED:
JHAMALI C. PANGCATAN
ENGINEER II

REVIEWED:
MARIA MONINA P. DELIZO
OIC, FLOOD CONTROL, SOCIAL AND
ENVIRONMENTAL SECTION
DATE:

SUBMITTED:
ALBERTO B. ESQUEJO, JR.
CHIEF, PLANNING AND DESIGN DIVISION
DATE:

RECOMMENDED:
MATHIAS L. MALENAB
ASST. REGIONAL DIRECTOR
DATE:

APPROVED:
RONNEL M. TAN, CESO III
REGIONAL DIRECTOR
DATE:

SET NO.:
SHEET NO.:
07
08

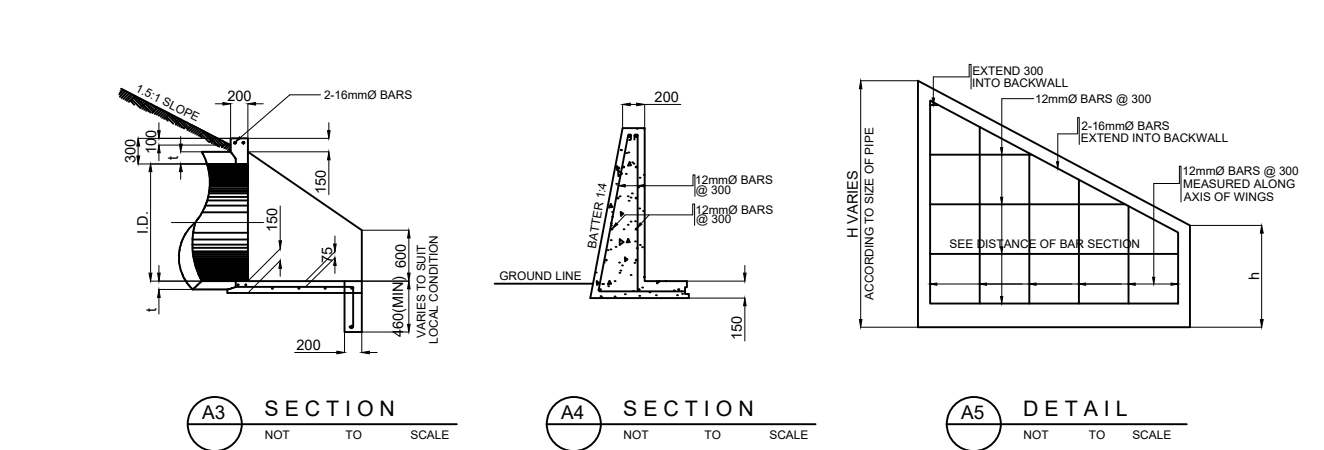
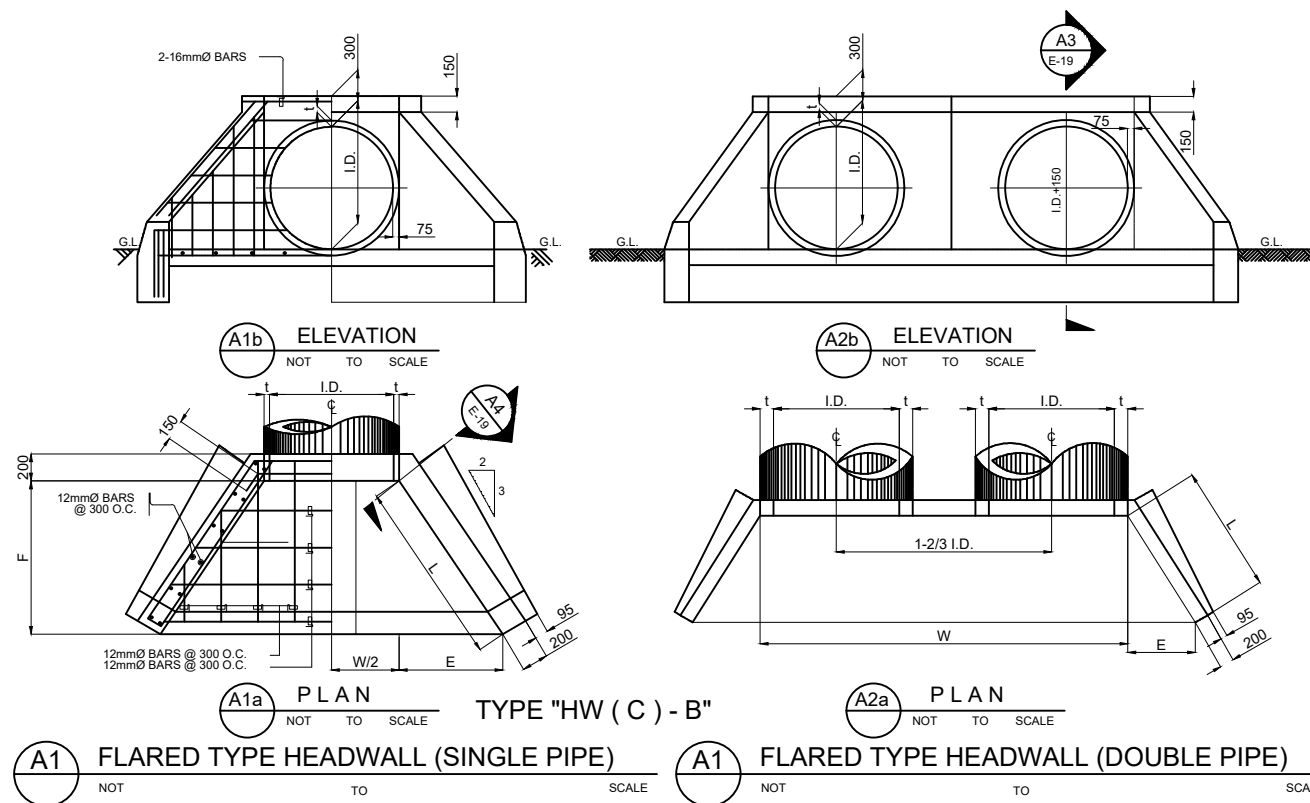


TABLE B (ONE FLARED TYPE HEADWALL 1.5:1)																	
DIA. & THICKNES(mm)		DIMENSIONS (mm)				SINGLE PIPE				DOUBLE PIPE				TRIPLE PIPE			
INTERNAL DIAMETER (I.D.)	MIN.THK. SHELL(t)	L	E	F	h	AREA OF WATERWAY m ³	W (mm)	EST. OF QUANTITIES		AREA OF WATERWAY m ²	W (mm)	EST. OF QUANTITIES		AREA OF WATERWAY m ²	W (mm)	EST. OF QUANTITIES	
								CONC. m ³	REINF-STEEL kg			CONC. m ³	REINF-STEEL kg			CONC. m ³	REINF-STEEL kg
460	50	690	380	570	0	0.17	610	0.55	12.70	0.33	1370	0.78	21.77	0.50	2100	1.05	46.55
610	60	960	530	800	0	0.29	760	0.82	15.87	0.58	1780	1.16	24.94	0.88	2770	1.55	69.19
910	80	1510	840	1260	600	0.65	1070	1.55	24.94	1.30	2590	2.22	47.62	1.95	4110	3.26	140.32
1070	90	1100	610	910	600	0.90	1220	1.45	61.22	1.80	3000	2.41	97.50	2.70	4820	4.16	177.11
1220	110	1370	760	1140	600	1.17	1370	1.88	83.45	2.34	3400	3.10	130.61	3.51	5510	5.15	218.26
1520	130	1920	1070	1600	600	1.81	1680	2.86	133.79	3.63	4220	4.71	200.45	5.44	6850	7.40	305.72

